THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Quadrant I	Check the signs of the abscissa and ordinate.
2	**(d)** 1	The degree is the highest power of the variable.
3	**(a)** $\sqrt{49}$	A rational number can be written as p/q with $q \neq 0$.
4	**(a)** $x^2 + 4x + 4$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(c)** 6	The abscissa is the x-coordinate.
6	**(c)** 11	Substitute x and simplify.
7	**(a)** $\sqrt{2}$	The square root of a non-square natural number is irrational.
8	**(a)** $x^2 - 8x + 16$	$(a-b)^2 = a^2 - 2ab + b^2$
9	**(b)** -6	The ordinate is the y-coordinate.
10	**(b)** 5	Solve $p(x) = 0$.
11	**(c)** 0.75	Divide the numerator by the denominator.
12	**(b)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
13	**(b)** 5	$\sqrt{3^2 + 4^2} = 5$
14	**(b)** 5	Put $x = 0$: $y = c$.
15	**(a)** $0.\overline{3}$	Divide the numerator by the denominator.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Abscissa $= 4$, ordinate $= 3$ (1); Point $= (4, 3)$ in Quadrant I (1).	1+1
17	$p(2) = 3(2) + (5)$ (1); $p(2) = 11$ (1).	1+1
18	Divide 3 by 5 (1); $\frac{3}{5} = 0.6$ (1).	1+1
19	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $= x^2 + 16x + 64$ (1).	1+1
20	$d = \sqrt{6^2 + 8^2} = \sqrt{100}$ (1); $d = 10$ units (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$AB = \sqrt{(4-(1))^2 + (6-(2))^2} = \sqrt{(3)^2 + (4)^2}$ (1); $AB = \sqrt{9+16} = \sqrt{25}$ (1); $AB = 5$ units (1).	1+1+1
22	$2x - 10 = 0$ gives $x = 5$ (1); Check: $p(5) = 2(5) + (-10) = 0$ (1); Hence the zero is verified (1).	1+1+1
23	Let $x = 0.\overline{2}$; $10x = 2.\overline{2}$ (1); $10x - x = 2$, so $9x = 2$ (1); $x = \frac{2}{9}$ (1).	1+1+1
24	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $a = 2x$, $b = 3$ (1); $= 4x^2 + 12x + 9$ (1).	1+1+1
	OR: Signs $(+, +)$: Quadrant I (1); Reflection in x-axis: $(5, -8)$ (1); Reflection in y-axis: $(-5, 8)$ (1).	1+1+1
25	$x = 0$: $y = 1$ (1); $x = 1$: $y = 4$ (1); $x = 2$: $y = 7$ (1).	1+1+1
	OR: Divide 5 by 12 (1); $\frac{5}{12} = 0.41\overline{6}$ (1); Repeats: block 6 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB = \sqrt{15^2 + 0^2} = 15$ (1); $BC = \sqrt{0^2 + 20^2} = 20$ (1); $AC = \sqrt{225 + 400} = 25$ (1); $AB^2 + BC^2 = 225 + 400 = 625$ (1); $AC^2 = 625$, so $AB^2 + BC^2 = AC^2$: right angle at B (1).	1+1+1+1+1
	OR: $F(x) = 10x + 30$ (1); $F(6) = 10 \times 6 + 30$ (1); $F(6) =$ ₹90 (1); $10x + 30 = 110$ (1); $x = 8$ km (1).	1+1+1+1+1
27	$\sqrt{25}$ is rational (1); $\sqrt{6}$ is irrational (1); $\sqrt{36}$ is rational (1); $\frac{3}{7}$ and 0.6 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $(2x)^2 = 4x^2$, $2(2x)(y) = 4xy$ (1); $= 4x^2 + 4xy + y^2$ (1); At $x = y = 1$: 9 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 961	$31 = (30 + 1)$, apply $(a + b)^2$
2	**(b)** $(4, 6)$	Average the abscissae and the ordinates.
3	**(d)** 3	In $y = mx + c$ the slope is m.
4	**(d)** terminating	Recall the basic facts.
5	**(d)** 2499	$(50 - 1)(50 + 1) = 2500 - 1$
6	**(a)** ordinate 0	Recall the properties of the axes.
7	**(a)** straight line	Recall the definition.
8	**(a)** 42	Apply Brahmagupta's rules for signed numbers.
9	**(b)** $(x + 5)(x - 5)$	$a^2 - b^2 = (a + b)(a - b)$
10	**(b)** $(3, -4)$	Only the sign of the ordinate changes.
11	**(a)** $3n + 2$	$a_n = a_1 + (n - 1)d$.
12	**(c)** 14	$14^2 = 196$
13	**(c)** 49	Add first, then square.
14	**(a)** 7	Same ordinate: distance $= \lvert x_2 - x_1 \rvert$.
15	**(d)** $(0, 4)$	At $x = 0$, $y = c$.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Put $3x - 15 = 0$ (1); $x = 5$ (1).	1+1
17	$\sqrt{16} = 4 = \frac{4}{1}$ (1); So it is rational (1).	1+1
18	Use $(a + b)(a - b) = a^2 - b^2$ (1); $= x^2 - 64$ (1).	1+1
19	$\left(\frac{2+6}{2}, \frac{10+4}{2}\right)$ (1); Midpoint $= (4, 7)$ (1).	1+1
20	Degree $= 1$ (1); The polynomial is linear (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 3$ (1); $= (x + 3)^2$ (1).	1+1+1
22	Midpoint $= (1, 2)$ (1); $d = \sqrt{(1)^2 + (2)^2}$ (1); $d = \sqrt{5}$ units (1).	1+1+1
23	$d = 4$ (1); $a_n = 4n - 1$ (1); $a_{10} = 39$ (1).	1+1+1
24	Add: $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$ (1); Halve it: $\frac{1}{2} \times \frac{3}{4}$ (1); A rational number between them is $\frac{3}{8}$ (1).	1+1+1
	OR: $= (50 + 5)(50 - 5)$ (1); $= 50^2 - 5^2 = 2500 - 25$ (1); $= 2475$ (1).	1+1+1
25	$AB = \sqrt{(3 - (-2))^2 + (13 - (1))^2} = \sqrt{(5)^2 + (12)^2}$ (1); $AB = \sqrt{25 + 144} = \sqrt{169}$ (1); $AB = 13$ units (1).	1+1+1
	OR: $3x + 12 = 0$ gives $x = -4$ (1); Check: $p(-4) = 3(-4) + (12) = 0$ (1); Hence the zero is verified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Quadrant: Quadrant I (1); Distance from x-axis $= \lvert 4 \rvert = 4$; from y-axis $= \lvert 3 \rvert = 3$ (1); $OP = \sqrt{9 + 16} = 5$ (1); Reflection in the origin: $(-3, -4)$ (1); The origin is the midpoint of P and its image: $\left(\frac{3+(-3)}{2}, \frac{4+(-4)}{2}\right) = (0, 0)$ (1).	1+1+1+1+1
	OR: Slope $= 2$ (1); y-intercept $= 6$ (1); x-intercept: $0 = 2x + (6)$, $x = -3$ (1); At $x = 3$: $y = 12$ (1); So the point lies on the line (1).	1+1+1+1+1
27	Long division gives remainders that repeat (1); $\frac{5}{11} = 0.\overline{45}$ (1); Repeating block 45 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1

	OR: Find two numbers with sum 5 and product 6 (1); The numbers are 2 and 3 (1); $= (x+2)(x+3)$ (1); Check by expanding (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{7}{9}$	A rational number can be written as p/q with $q \neq 0$.
2	**(d)** $x^2 + 5x + 6$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
3	**(c)** Quadrant II	Check the signs of the abscissa and ordinate.
4	**(d)** 3	The degree is the highest power of the variable.
5	**(c)** $\sqrt{3}$	The square root of a non-square natural number is irrational.
6	**(a)** $x^2 + 6x + 9$	$(a + b)^2 = a^2 + 2ab + b^2$
7	**(d)** -7	The abscissa is the x-coordinate.
8	**(a)** 14	Substitute x and simplify.
9	**(b)** 0.875	Divide the numerator by the denominator.
10	**(b)** $x^2 - 10x + 25$	$(a - b)^2 = a^2 - 2ab + b^2$
11	**(a)** 7	The ordinate is the y-coordinate.
12	**(c)** 3	Solve $p(x) = 0$.
13	**(a)** $0.\overline{6}$	Divide the numerator by the denominator.
14	**(d)** $x^2 - 16$	$(a + b)(a - b) = a^2 - b^2$
15	**(a)** 13	$\sqrt{5^2 + 12^2} = 13$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Use the sign rule for $(-9) \times (-4)$ (1); $= 36$ (1).	1+1
17	$42 = (40 + 2)$ (1); $42^2 = 1600 + 160 + 4 = 1764$ (1).	1+1
18	Abscissa $= -5$, ordinate $= 2$ (1); Point $= (-5, 2)$ in Quadrant II (1).	1+1
19	$p(5) = 4(5) - 3$ (1); $p(5) = 17$ (1).	1+1
20	Divide 5 by 8 (1); $\frac{5}{8} = 0.625$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Let $x = 0.\overline{4}$; $10x = 4.\overline{4}$ (1); $10x - x = 4$, so $9x = 4$ (1); $x = \frac{4}{9}$ (1).	1+1+1
22	Use $(a + b)^2 = a^2 + 2ab + b^2$ (1); $a = 3x$, $b = 1$ (1); $= 9x^2 + 6x + 1$ (1).	1+1+1
23	Signs $(-, +)$: Quadrant II (1); Reflection in x-axis: $(-4, -6)$ (1); Reflection in y-axis: $(4, 6)$ (1).	1+1+1
24	$x = 0$: $y = 5$ (1); $x = 1$: $y = 7$ (1); $x = 2$: $y = 9$ (1).	1+1+1
	OR: Divide 7 by 15 (1); $\frac{7}{15} = 0.4\overline{6}$ (1); Repeats: block 6 (1).	1+1+1
25	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 4$ (1); $= (x + 4)^2$ (1).	1+1+1
	OR: Midpoint $= (4, 2)$ (1); $d = \sqrt{(4)^2 + (2)^2}$ (1); $d = 2\sqrt{5}$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB = \sqrt{10^2 + 0^2} = 10$ (1); $BC = \sqrt{0^2 + 24^2} = 24$ (1); $AC = \sqrt{100 + 576} = 26$ (1); $AB^2 + BC^2 = 100 + 576 = 676$ (1); $AC^2 = 676$, so $AB^2 + BC^2 = AC^2$: right angle at B (1).	1+1+1+1+1
	OR: $F(x) = 12x + 20$ (1); $F(5) = 12 \times 5 + 20$ (1); $F(5) =$ ₹80 (1); $12x + 20 = 104$ (1); $x = 7$ km (1).	1+1+1+1+1
27	$\sqrt{2}$ is irrational (1); $\sqrt{49}$ is rational (1); $\sqrt{64}$ is rational (1); $\frac{5}{9}$ and 0.45 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Use $(a + b)^2 = a^2 + 2ab + b^2$ (1); $(3x)^2 = 9x^2$, $2(3x)(2y) = 12xy$ (1); $= 9x^2 + 12xy + 4y^2$ (1); At $x = y = 1$: 25 (2).	1+1+1+2

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